

2assignment newton2

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:33pm

Quiz Instructions

No tutorial but hints. HIDE the hints first.

For every action there is an equal and opposite reaction HOWEVER the consequence of the force depends on the mass. (consequence is the acceleration).



Question 1 4.5 pts

Two blocks (A and B) are in contact on a horizontal frictionless surface. A 20 N constant horizontal force is applied to A. The mass of A is 2.0 kg and the mass of B is 6.0 kg. The magnitude of the force of B on A is:

- a. 2.5 N.
- b. 15 N.
- c. 10 N.
- d. 5.0 N.
- e. 20 N.



I did in class. hide hint

hint: its easier to find the force applied of A on B. Since its the same as B on A. system B has only one force applied to it. First find the acceleration using the 2 blocks as a system. as done in class. then use only block B as your system. its the same acceleration

☐

d

☐

e

☐

a

☐

b



c



Question 2 4.5 pts

Joe's weight, to two significant figures, is 700 newtons down. What is the magnitude of the total force he needs to apply in order to accelerate at 2.0 m.s^{-2} ?

Magnitude of required force to accelerate Joe = _____ N.

Hide the hint

Hint: don't over thing it. Find the mass of Joe. And what is force applied to Joe so his acceleration is 2m/s/s



1400 N



700N



350N



140N



Question 3 4.5 pts

A bicycle is travelling at constant velocity on a horizontal road

what is true ?

hint: constant velocity means?



the total force on the bike is antiparallel to the motion



various forces are acting on the bike and their sum is 0



the total force on the bike is parallel to the motion



no force is acting on the bike. acceleration is 0



Question 4 4.5 pts

When you walk, you push on the floor to the left and the floor



also pushes on you to the left.



pushes you to the right.



can only wish it could push on you.



none of the above



both of these simultaneously.



Question 5 4.5 pts

Suzie Skydiver, who weighs 500 N, reaches terminal velocity of 90 km/h. The air resistance on Suzie is then



410 N.



90 N.



250 N.



500 N.



none of the above



Question 6 4.5 pts

An object that weighs 75 N is pulled on a horizontal surface by a horizontal pull of 50 N to the right. The friction force on this object is 30 N to the left. What is the acceleration of the object?

hint: right - left = mass x acceleration



0.27 m/s^2



1.1 m/s^2

11 m/s²2.6 m/s²

Question 7 4.5 pts

A 1200-kg car is pulling a 500-kg trailer along level ground. Friction of the road on the trailer is negligible. The car accelerates with an acceleration of 1.3 m/s². What is the force exerted by the car on the trailer?

hint: right - left = mass x acceleration



700 N



600 N



650 N



550 N



750 N



Question 8 4.5 pts

When a cannonball is fired from a cannon, which undergoes the greater acceleration?

hint: Remember $Ma = \text{Force} = -m A$ (see slide)

force is the same. The consequence (change in the state of motion) depends on the mass.



the recoiling cannon



both the same



the cannonball



Question 9 4.5 pts

If I weigh 741 N on Earth at a place where $g = 9.80 \text{ m/s}^2$ and 5320 N on the surface of another planet, what is the acceleration due to gravity on that planet?

hint: weight = mass x (acceleration on planet)

61.2 m/s²81.0 m/s²51.4 m/s²70.4 m/s²

Question 10 4.4 pts

A sack of potatoes weighing 200 N falls from an airplane. As the velocity of fall increases, air resistance also increases. When air resistance equals 200 N, the sack's acceleration becomes

0 m/s².10 m/s².

infinite.

5 m/s².

none of the above



Question 11 4.4 pts

A force of 1 N accelerates 1-kg box at the rate of 1 m/s². The acceleration of a 2-kg box by a net force of 2 N is



half as much.



twice as much.



none of the above



the same.



Question 12 4.4 pts

A tow truck exerts a force of 3000 N on a car, which then accelerates at 2 m/s². What is the mass of the car?



3000 kg

- ☐ 1500 kg
- ☐ 500 kg
- ☐ 1000 kg
- ☐ none of these



Question 13 4.4 pts

Nellie pulls on a 10-kg wagon with a constant horizontal force of 30 N. If there are no other horizontal forces, what is the wagon's acceleration?

- ☐ 0.3 m/s²
- ☐ 30 m/s²
- ☐ 3.0 m/s²
- ☐ 300 m/s²
- ☐ 10 m/s²



Question 14 4.4 pts

An astronaut of mass 70 kg weighs 700 N on Earth's surface. His weight on the surface of Mars, where the acceleration due to gravity is 3.7 m/s², would be about

- ☐ none of the above
- ☐ 260 N.
- ☐ 370 N.
- ☐ the same as on Earth.
- ☐ 130 N.



Question 15 4.4 pts

A vertical vector of 3 units combined with a horizontal vector of 4 units has a resultant of

- ☐ 7 units.



1 unit.



5 units.



Question 16 4.4 pts

You stand on your skateboard and exert a 50-N push on the wall next to you. If your mass is 60 kg you'll momentarily accelerate from the wall at about



none of the above



0.08 m/s².



8.0 m/s².



0.8 m/s².



Question 17 4.4 pts

An archer shoots an arrow. Consider the action force to be the bowstring against the arrow. The reaction to this force is the



combined weight of the arrow and bowstring.



friction of the ground against the archer's feet.



grip of the archer's hand on the bow.



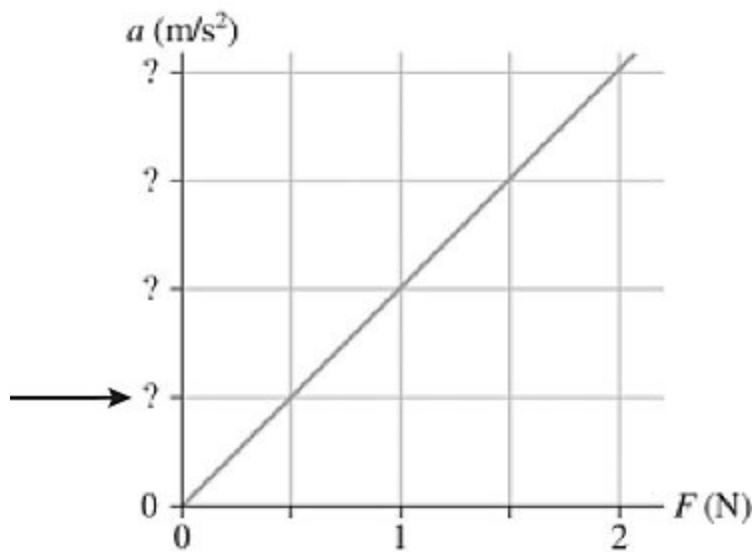
arrow's push against the bowstring.



Question 18 4.4 pts

The figure shows an acceleration-versus-force graph for a 125-g object. What should be the value of the first tick-mark on the vertical scale, as indicated by the arrow in the figure?

HIDE the HINT first.



hint: according to picture, The force applied is 0.5N

☐

4

☐

8

☐

0.00400

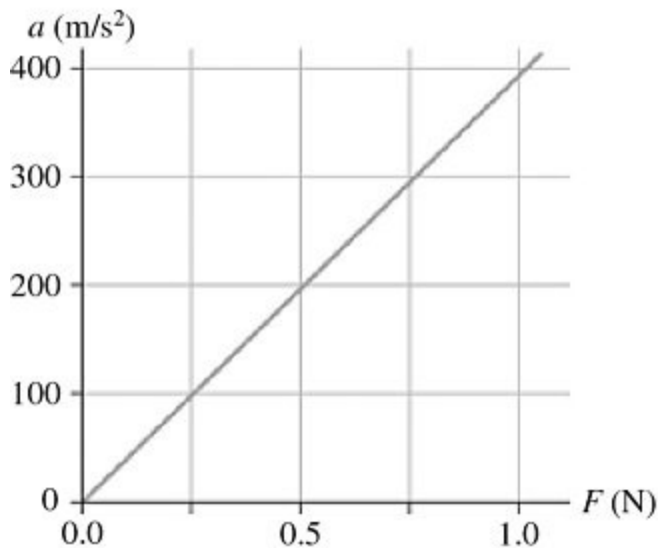
☐

0.00800

⋮

Question 19 4.4 pts

The figure shows an object's acceleration-versus-force graph. What is the mass of this object?



hint: what is the relationship between a and F ? (newton's second law).

Whats a slope?

☐

2.5 g

☐

1.6 g

☐

400,000 g

☐

630 g

☐

Question 20 4.4 pts

Calculate the average force a bumper would have to exert to bring a 1200-kg car to rest in 15 cm when the car had an initial speed of 2.0 m/s (about 4.5 mph).

hint: first use kinematics to get the acceleration. With $V_f=0$. convert to meters the horizontal displacement Δx

☐

$1.6 \times 10^4 \text{ N}$

☐

$5.4 \times 10^4 \text{ N}$

☐

$1.8 \times 10^4 \text{ N}$

☐

$3.2 \times 10^4 \text{ N}$

☐

$6.5 \times 10^5 \text{ N}$

☐

Question 21 4.5 pts

When a rocket forces exhaust gases downward, the exhaust gases

☐

as the name implies, soon exhaust their influence.

☐

exert an upward force on the rocket.

☐

play no further role in the motion of the rocket.



Question 22 4.4 pts

Action and reaction pairs of forces

☐

are independent of time.

☐

always act simultaneously.

☐

may or may not act simultaneously.



Question 23 4.5 pts

Arnold Strongman and Suzie Small each pull very hard on opposite ends of a rope in a tug-of-war. The greater force on the rope is exerted by

☐

Arnold, of course.

☐

both the same, interestingly.

☐

Suzie, surprisingly.



Question 24 3 pts

your weight is 800N . thats your real weight.

if you stand on a 40 degrees slope. your weight is

hint: whats the normal force

☐

800N

☐

514N

☐

613N

☐

0N

Not saved

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